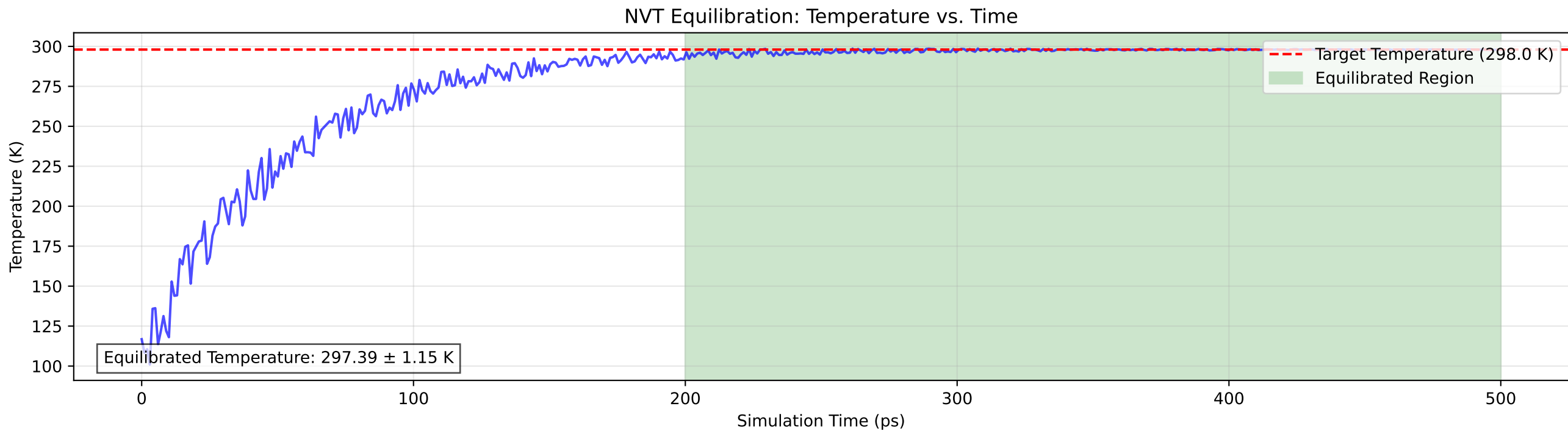


NVT Equilibration in Molecular Dynamics:

- Purpose:** To bring the system to the target temperature before production runs
- Key Characteristics:**
 - N (Number of particles): Fixed
 - V (Volume): Constant
 - T (Temperature): Controlled by a thermostat
- Process Steps:**
 - Initial velocities are assigned from a Maxwell-Boltzmann distribution
 - Thermostat (e.g., V-rescale or Nosé-Hoover) adjusts velocities to reach target temperature
 - Temperature equilibrates with natural fluctuations around the target value
 - Energy fluctuations become more regular as the system reaches thermal equilibrium
- Thermostats:**
 - V-rescale: Velocity rescaling with a stochastic term (efficient temperature control)
 - Nosé-Hoover: More accurately samples the canonical ensemble, but slower to equilibrate
 - Berendsen: Simple but distorts the ensemble; useful for initial equilibration
- Next Steps after NVT:**
 - NPT equilibration to stabilize pressure and density
 - Production MD with proper ensemble sampling